# Niklas Schlomberg

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## Education

Nov 2020 - Apr 2026 **PhD in Discrete Mathematics**, University of Bonn, Germany
Thesis: *Cycle Packing and Cycle Transversal in Graphs on Oriented Surfaces*
Advisors: Prof. Dr. Jens Vygen and Prof. Dr. László Végh
Final grade: 0.7 (magna cum laude)

Oct 2018 - Oct 2020 **M.Sc. in Mathematics**, University of Bonn, Germany
Thesis: *Scheduling and Vehicle Routing*
Advisors: Prof. Dr. Jens Vygen and Prof. Dr. Stephan Held
Final grade: 1.1 (very good)

Oct 2015 – Sept 2018 **B.Sc in Mathematics**, University of Bonn, Germany
Thesis: *TSP mit dynamischer Programmierung*
Advisors: Prof. Dr. Jens Vygen and Prof. Dr. Stephan Held
Final grade: 1.0 (excellent)

## Work experience

Since May 2026 **Postdoctoral researcher**, *IDSIA, USI-SUPSI, Lugano*, Switzerland
in the group of Fabrizio Grandoni

Nov 2020 - Apr 2026 **Research Assistant**, *Research Institute for Discrete Mathematics, University of Bonn*, Germany
Cooperation with IBM on Detailed Routing in Chip Design

Mar 2018 - Oct 2020 **Student Research Assistant**, *Research Institute for Discrete Mathematics, University of Bonn*, Germany
Cooperation with the German Post on Vehicle Routing

## Distinctions

2018 **Bachelor Prize of the Bonn Mathematical Society (BMG)**, *University of Bonn*

2015 **Reached final round of the national mathematics competition in Germany**

## Teaching activities

Summer term, 2021 **Teaching assistant**, *Lecture "Chip Design"*

Summer term, 2024 **Teaching assistant**, *Lecture "Chip Design"*

2020 – 2025 Co-supervised several Master's and Bachelor's theses

2020 – 2026 Co-supervised several applied student projects on Chip Design

## Applied projects

- **Vehicle Routing**
  - Developed / implemented an exact algorithm for TSP with time-dependent travel times
  - Developed / implemented algorithms for robust scheduling with breaks
- **Chip Design**
  - Developed / implemented various heuristics for timing-aware Detailed Routing

## Conference participations and Talks

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|---|---|
| July 7 – 10, 2026 | **International Colloquium on Automata, Languages, and Programming (ICALP) 2026**, *Royal Holloway, UK*, Talk titled "Pinning on Tight Cuts: Improved Algorithm and Bounds for Unsplittable Multicommodity Flows in Outerplanar Graphs" |
| June 17 – 19, 2026 | **IPCO 2026**, University of Padova, Italy |
| Apr 20 – 24, 2026 | **Follow-Up Workshop: "Discrete Optimization"**, *University of Bonn, Germany*, Talk titled "Improved Erdős–Pósa inequalities for odd cycles in planar graphs" |
| Dec 3, 2025 | Invited talk at IDSIA, USI-SUPSI, Lugano |
| Sept 2, 2024 | Invited talk at the FOC seminar of Masaryk University Brno |
| July 21 – 26, 2024 | **International Symposium on Mathematical Programming (ISMP) 2024**, *Montreal, Canada*, Contributed talk |
| July 8 – 12, 2024 | **International Colloquium on Automata, Languages, and Programming (ICALP) 2024**, *Tallinn, Estonia*, Talk titled "An improved integrality gap for disjoint cycles in planar graphs" |
| June 2 – 4, 2023 | **Highlights of Algorithms (HALG) 2023**, *Prague, Czech Republic*, Contributed Talk |
| Jan 22 – 25, 2023 | **Symposium on Discrete Algorithms (SODA) 2023**, *Florence, Italy*, Talk titled "Packing cycles in planar and bounded-genus graphs" |
| Oct 17 – 21, 2022 | **Bonn Workshop on Combinatorial Optimization**, *University of Bonn*, Germany, Invited Talk |
| Sept 19 – 23, 2022 | **11th Cargese Workshop on Combinatorial Optimization**, *Institut d'Etudes Scientifiques de Cargèse*, Corsica, France |
| June 27 – 29, 2022 | **IPCO 2022 Summer School**, Eindhoven University of Technology, Netherlands |

## Service/Reviewing Activities

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| Journal Reviewer | Transactions on Algorithms, Information Processing Letters |
| Conference Reviewer | FOCS, SODA, ICALP, IPCO, ESA, APPROX, ICCAD |